Meeting Minutes: NYU FRL Weekly Meeting

Saurabh and Ray – Friday 3rd April

Updates:

Saurabh:

Remove feature:

A newer, smarter, much spatially aware, and not something that "just removes" objects from the scene.

Before:

- No spatial / linguistic disambiguation (e.g. "chair by the table" → often all name matches).
- Plural/singular and "all X" vs "all objects" handled by brittle string/DB heuristics in AssetAgent + DB
- AssetAgent owned final IDs despite being library/name-oriented, not scene-layout-oriented, causing it to completely ignore spatial concepts like "remove chair to the right of the table"
- Graph hygiene: upstream failure could still hit Scene in some cases; verification valid was ignored before execution, so bad verification didn't reliably stop the run.

After:

- Now SceneAgent is the spatial owner for delete. The final id(s) comes from resolve_removal_targets(), parallel in role to placement for add/move.
- AssetAgent is intent-only for remove. It no longer makes the weak name-based search, instead emits remove_intent + scope_hint; does not finalize target_object_ids.
- Orchestrator always runs remove through Scene after Asset, merging remove_intent into the dict passed to removal resolution. This is in policy with ADD/ROT/POS, being true to the proposed execution heuristic
- VerificationAgent owns remove policy: validate_removal (existence, movable/structural, singular vs plural).
- Execution/broadcast: same delete + object_removed idea; payload can include name for logs/UI
- Safety: structural / non-movable deletes blocked in verification (wall, Floor, non-existent objects such as "remove the unicorn")

STT Feature final Integration:

Added an STT Feature that listens to and cleans up the Transcription, as discussed in some of the previous meetings.

Architecture (high level)

- Browser handles capture + first-pass speech-to-text (Web Speech in Chromium); backend handles LLM cleanup only, on a separate route from the main MAS command pipeline.
- STT is optional and isolated: natural-language execution still goes through the existing Execute → scene command path; nothing in the agent graph depends on voice.
- Frontend delivery keeps sttListen.js as its own module next to the webpack bundle so the dev server actually serves it.
- Check out ./stt for the code related to STT

Ray

Image Agent:

Ray Added the ImageAgent that gets the spatial representation in a SceneGraph and has pushed to a GitHub Branch. Saurabh will pull that branch, try to play around; merge the current two latest versions, and push it back to a new branch

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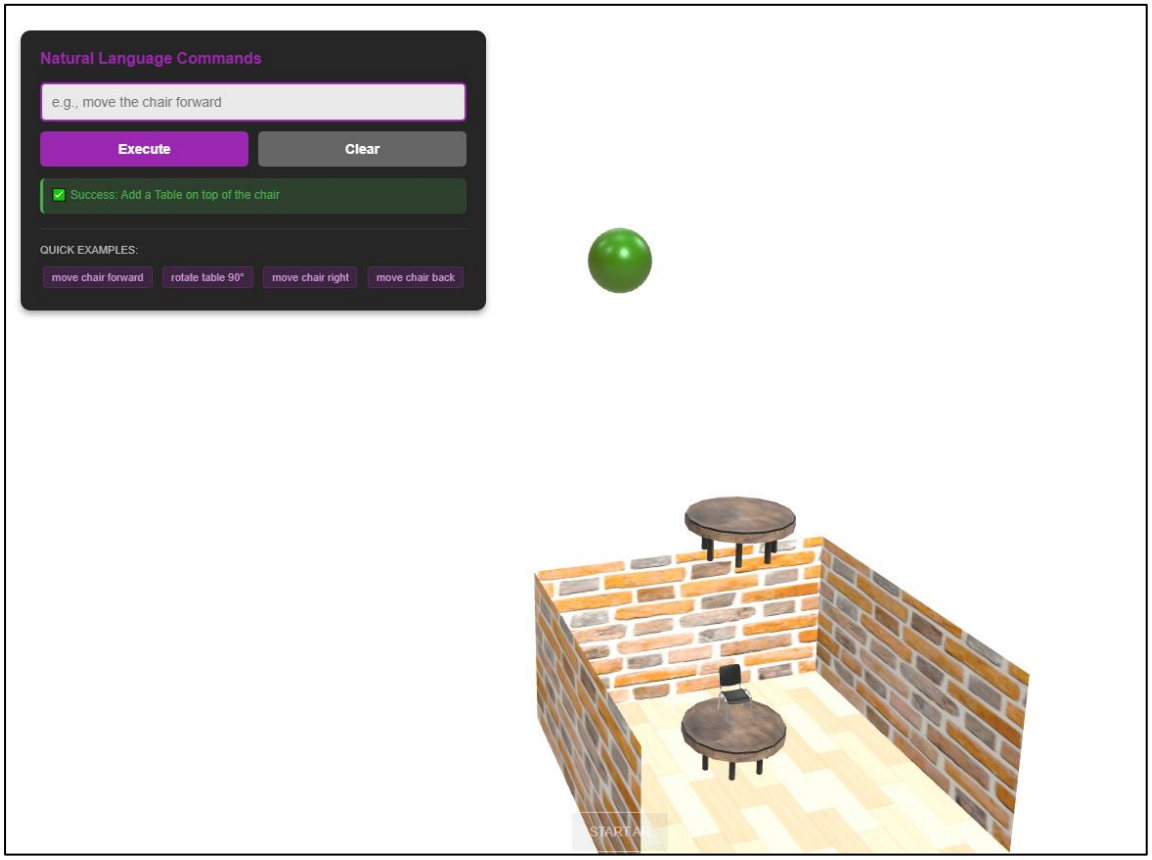
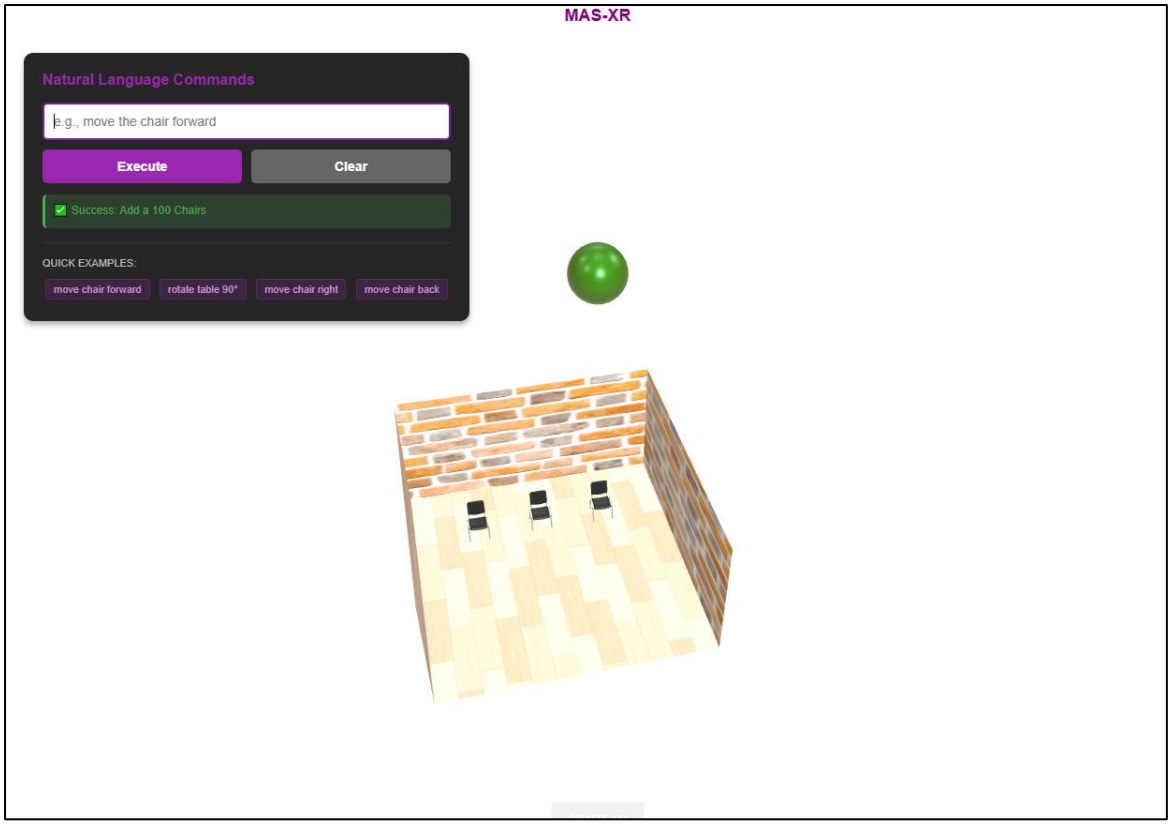
What we Discussed:

Saurabh showcased some basic test cases with the remove feature. It mostly involved remove commands spatial concepts such as “to the left of” etc. Most test cases worked fine as expected of the standard routing similar to ADD/POS/ROT. However, Saurabh also found many edge case failures as noted below:

1. The command “add a chair on top of the table and again add a table on top of the chair”, should ideally be an invalid action (a 4-legged Table cannot be on top of a Chair). Yet on top of it executing at all, it executes the action with a weird Z-Offset.

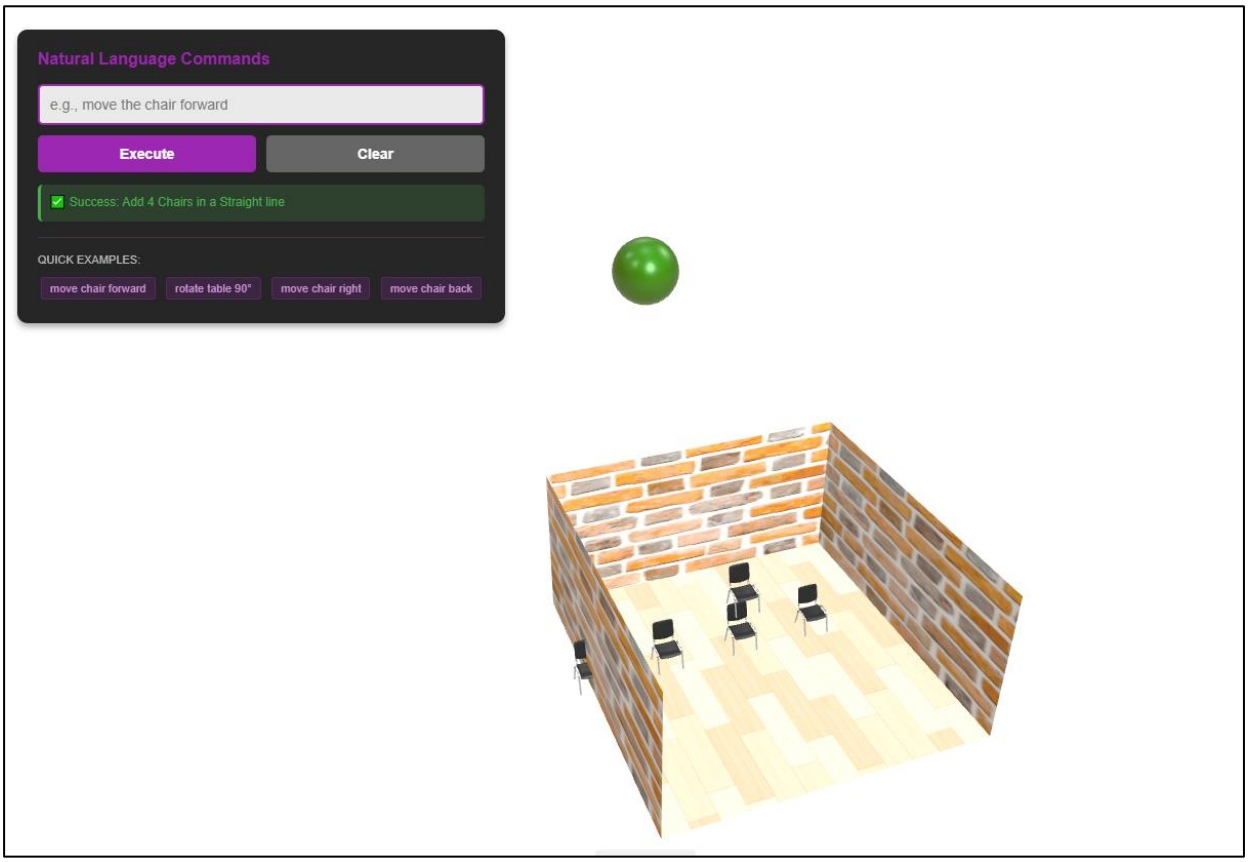


2. The command “Add a 100 Chairs” DOES NOT in fact add 100 chairs. This might be due to the direct regex of “a/an” being counted as “1”. Additionally, whether the system can add 100 chairs without malfunctioning due to the sheer size of the command.



3. This command of “remove the floor” works correctly, which is good. But the vague error message might be something that we need to work on. Such as “Why” did the error occur? (In this case it was invalid deletion on a key environment object, but the user doesn’t know that.)

4. This image shows two commands used in conjunction, and multiple spatial failures.
    a. We did ➔ “Add a table” ➔ “Add a Chair on top of the table” ➔ “Remove the table”. What happens once the table is removed successfully? We now see a floating chair! This should be an invalid action, yet because the system doesn’t really understand the relationship of objects, especially in the Z-Offset, it removes things indiscriminately
    b. Next, “Add four chairs in a straight line” ➔ Causes one of the chairs to clip with the left wall. This might be happening because The VerificationNode doesn’t check for collision with the walls themselves



5. Following case of “Add a table” ➔ “Add four chairs surrounding the take with inward facing orientation” is a fascinating use case of how we could encode the “orientation factor” into the SceneAgent. We know what the “front” part of the chair is. But does the LLM know that? It would - if we give it the correct context, at the cost of added complexity

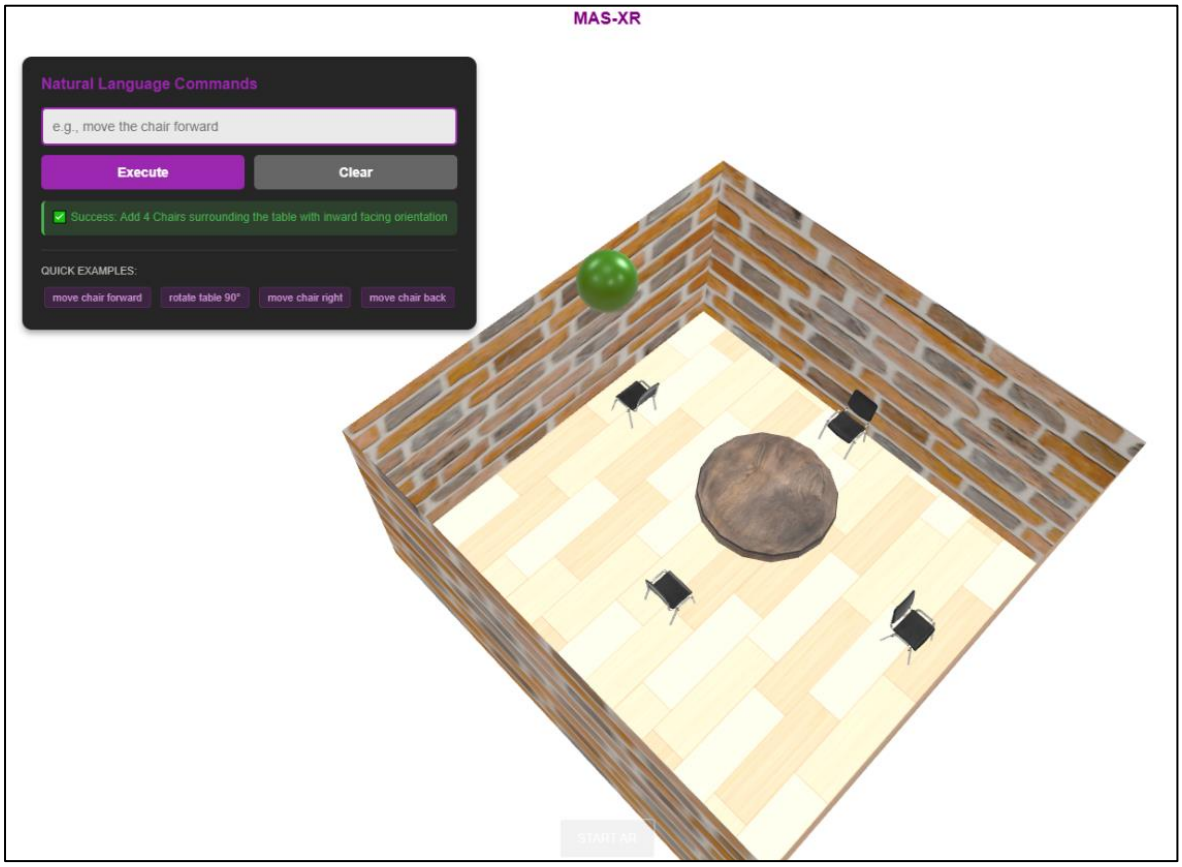


Image Agent - Ray added the Image Agent feature. He will add more related to that in the meeting note Blog

Literature Reviews and which architecture for a Spatial representation we can choose:

- Ray thinks the following two articles best suit our research goal
    - Articles One: 3D Dynamic Scene Graphs: Actionable Spatial Perception with Places, Objects, and Humans
    - Article Four: Translating Universal Scene Descriptions into Knowledge Graphs for Robotic Environment
- The other two articles might help us understand what we're dealing with from a more classic CV/Image Recognition, Object Detection perspective

User Studies Perspective: We discussed some things that need to be added from the perspective of the user studies, and facilitating the users, one such example is below:

- UI/ UX Perspective: what about feedback to the user? "Error: Failed" is not that helpful when user is trying something new. Although this is not research oriented, it might help us get better survey results in the User Studies that we create

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To-Do:

- Saurabh will commit a Branch of his own, which contains the STT + remove Feature.
- He will merge it with Ray's current branch after basic testing on both branches.

- He will then test it on his Windows machine (since Windows has more issues with programming and other dependencies as compared to Linux/mac).
- After confirmation of non-failure and some basic Test cases later, Saurabh Will commit to another *FINAL* branch, which ensures that Remove + STT + imageAgent_v1 are perfectly integrated
- Possibly try to solve the various spatial issues pertaining to Add and Remove.